# Embedded AI-Driven Single-Electrode Soft Multimodal Sensor for Proprioception, Tactile, and Proximity Sensing

Bio-Robotics and Control (BiRC) Lab
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Korea University

HeeJun Moon

# 구조

**1. Introduction**
- Needs for Multimodal Sensor →
- Limitations of Current Researches →

**2. Mechanism and Design of Multimodal Sensor**
- Sensor Design →
- Sensor Modal Characteristics →
- TDM Mechanism →

**3. Fabrication (Test Setup)**
- Sensor Fabrication → (Q-factor 고려 프린팅)
- PCB Fabrication →
- Test Stage Fabrication →

**4. NN-Model Learning**
- Model Structure →
- Data Acquisition → 1. 데이터 취득 동영상 2. 데이터 그래프(2D:인장,근접,압력/3D:인장+근접,인장+
- Learning Result → 1. RMSE

**5. Realtime-EI Multimodal Decoupling**
- Embedding →
- Realtime Decoupling Result → 1. RMSE, 2. Latency, 3.

**6. Future Work**
- Application → 1. Research Design 슬라이드

# 구조

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1.2 Limitations of Current Research
1.3 Purpose and Significance of This Study
**2. Sensor Design and Fabrication**
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2.3 Sensor Fabrication
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5.2 Embedded Implementation
5.3 Real-Time Validation
**6. Discussion  7. Conclusion** 요약 + Future Work

# I. Introduction

# I. Introduction

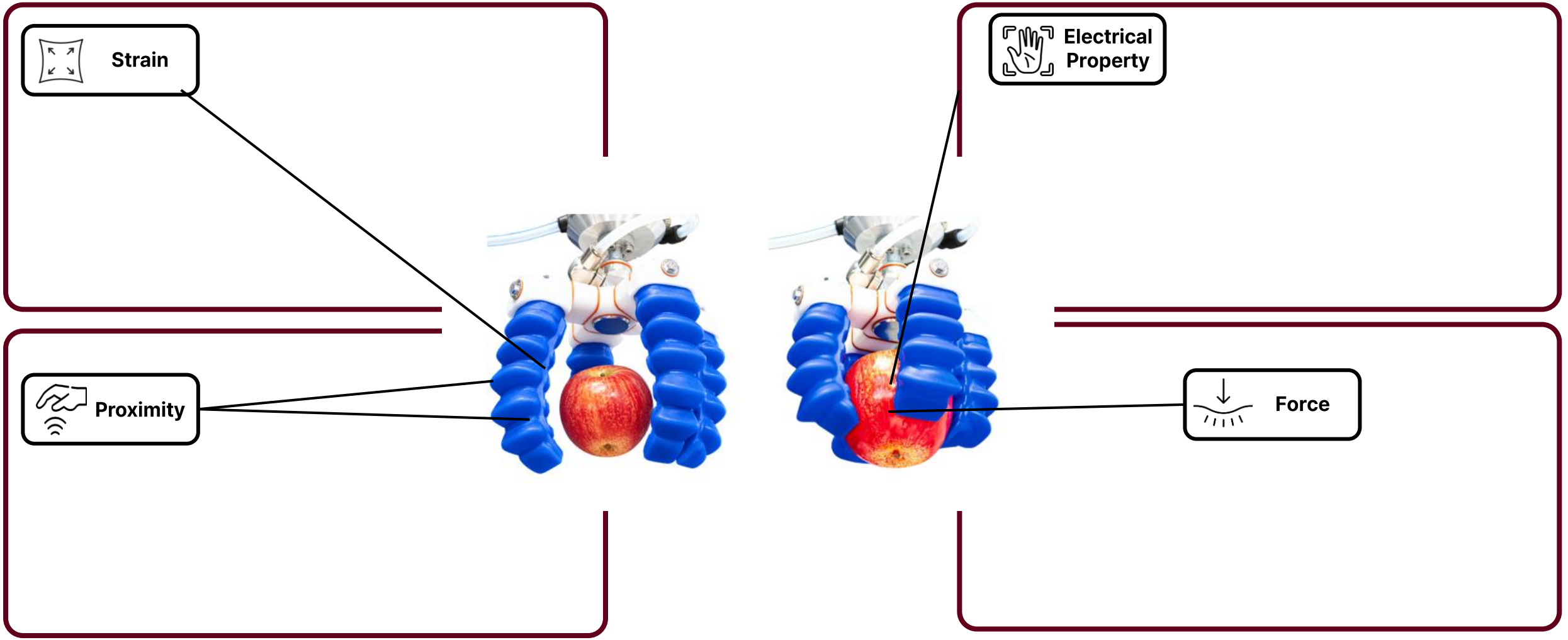
## 1.1 Background

HMI, Soft Robot



# I. Introduction

## 1.1 Background



# I. Introduction

## 1.2 Limitations of Current Research

**Fundamental Challenge**

Soft Multimodal Sensor의 한계점들
- 박리
- 배선
- 등등

[5] "Multimodal sensors with decoupled sensing mechanisms." Advanced Science 9.26 (2022)

# I. Introduction

## 1.2 Limitations of Current Research

**Fundamental Challenge**

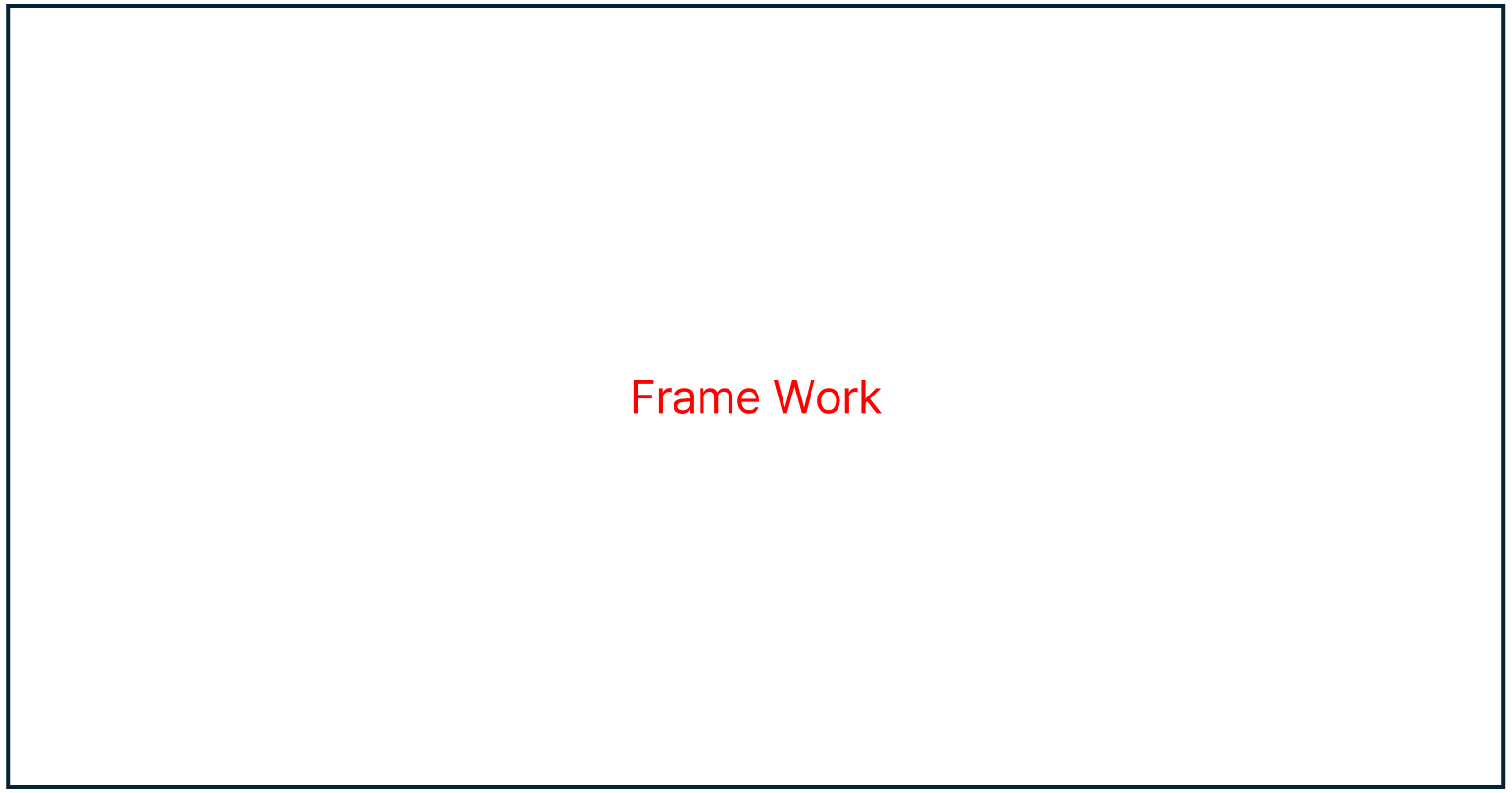
현재 디커플링하는 Soft Multimodal 연구들의 한계

[5] "Multimodal sensors with decoupled sensing mechanisms." Advanced Science 9.26 (2022)

# I. Introduction

## 1.3 This Work

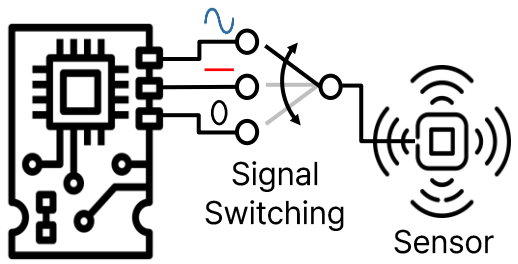
Onboard TDM
+ EI



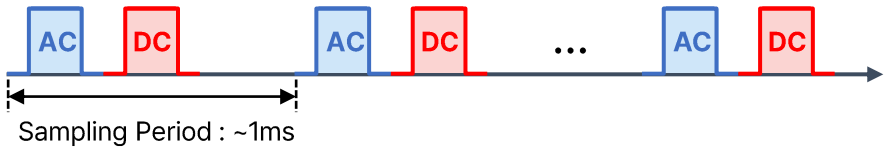
# I. Introduction

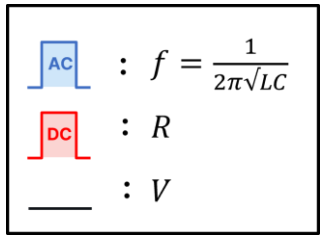
## 1.3 This Work

### Time Division Measurement (TDM)



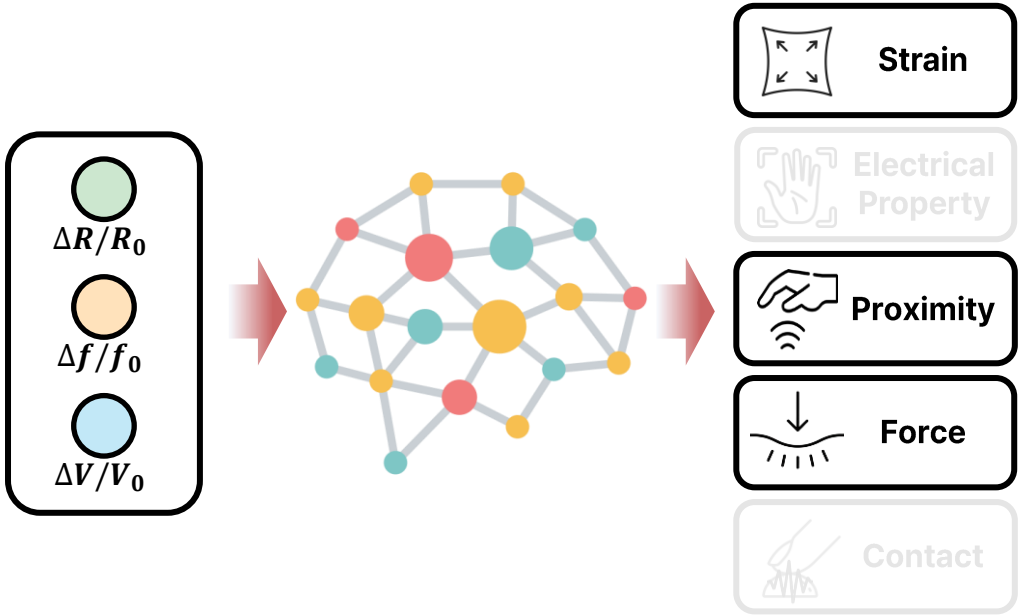

<Signal Acquisition Timeline>




- 시분할 측정 방식을 통해 단일 센서에 순차적으로 AC excitation, DC excitation, No excitation을 반복하여 인가
- 각 상태에서 센서의 **공진주파수**, **저항**, **전압**(TENG)을 각각 독립적으로 측정

### Modality Decoupling

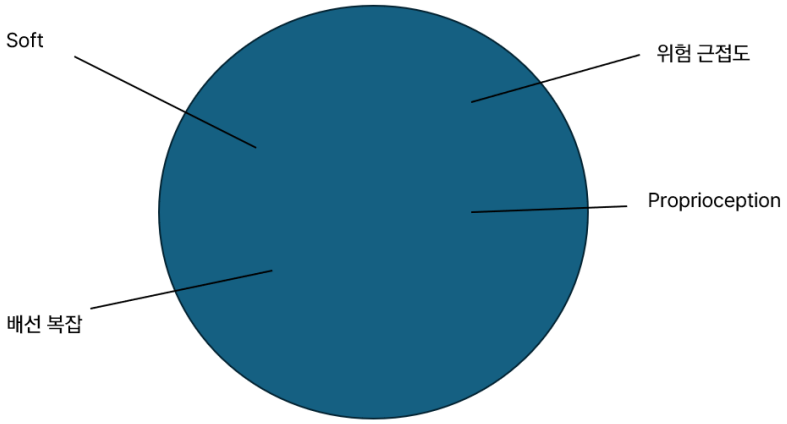



- 시분할 측정 방식으로 얻은 독립적인 물리량들을 학습모델로 디커플링하여 다중 모달(인장, 압력, 근접도, 물체의 전기적 성질, 촉감) 센싱

# I. Introduction

## 1.3 This Work

어떻게 극복하는지?




Chapter2

Chapter2

Chapter2

Chapter2

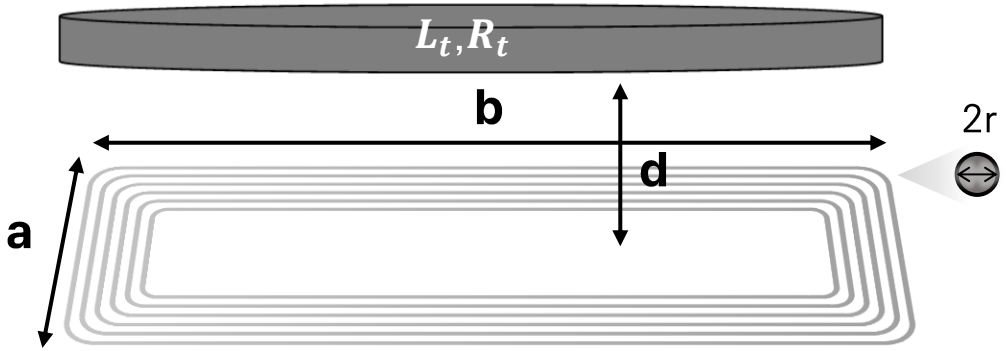
Chapter3

Chapter3

# II. Sensor Design and Fabrication

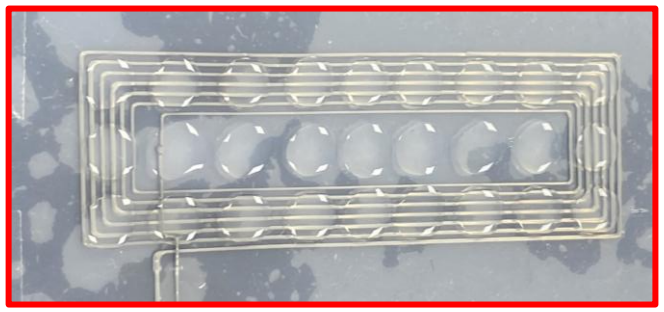
# II. Sensor Design and Fabrication

## 2.1 Sensor Design

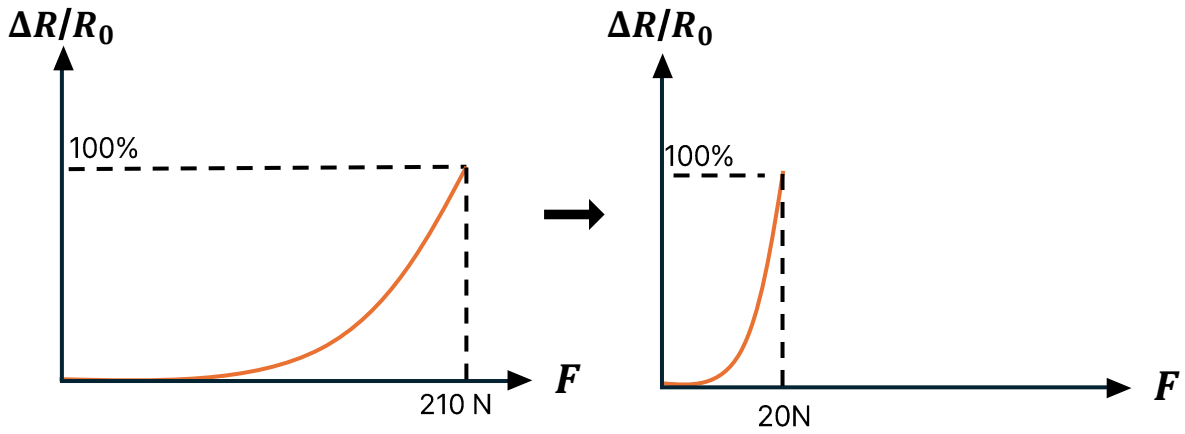



ble

chable



돌기구조 효과 언급



# II. Sensor Design and Fabrication

## 2.2 Mechanical and Electromagnetic Characteristics

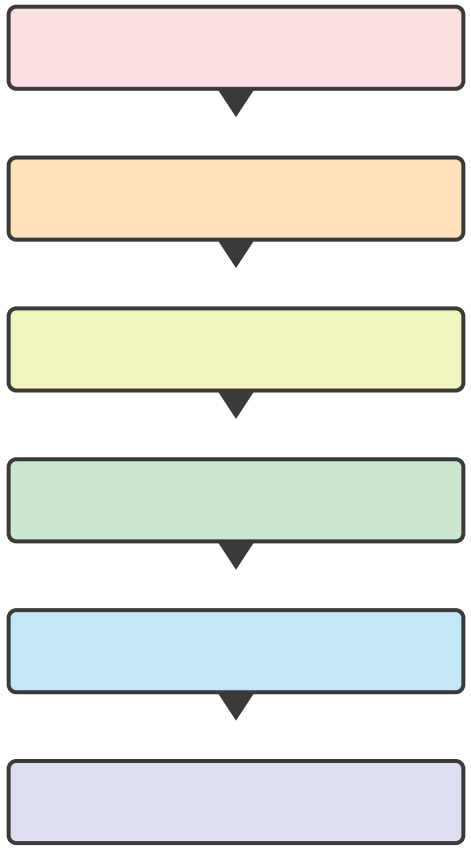
# II. Sensor Design and Fabrication

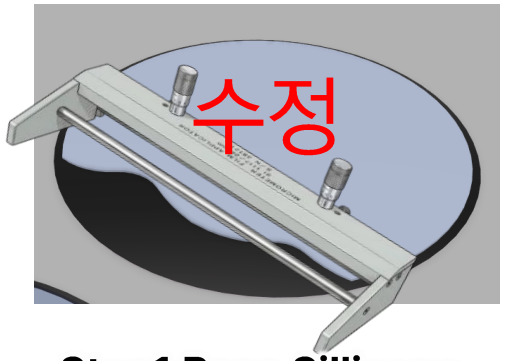
## 2.2 Mechanical and Electromagnetic Characteristics

# II. Sensor Design and Fabrication

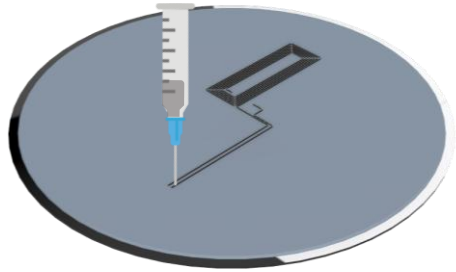
## 2.3 Sensor Fabrication

### (b)Fabrication Process

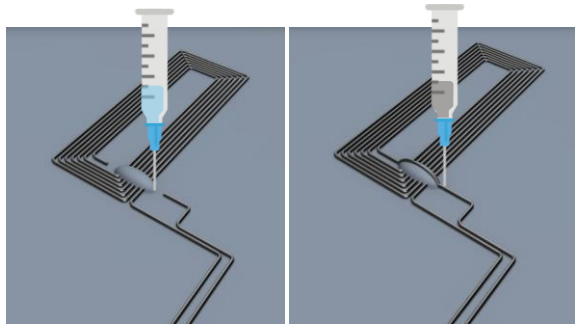




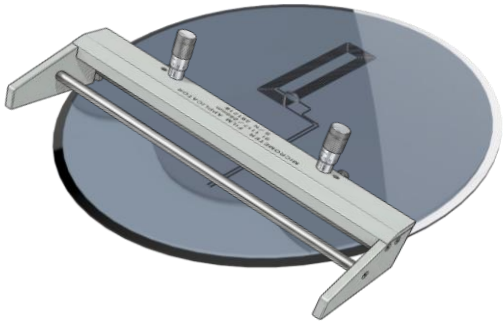

Step1 Base Sillicone



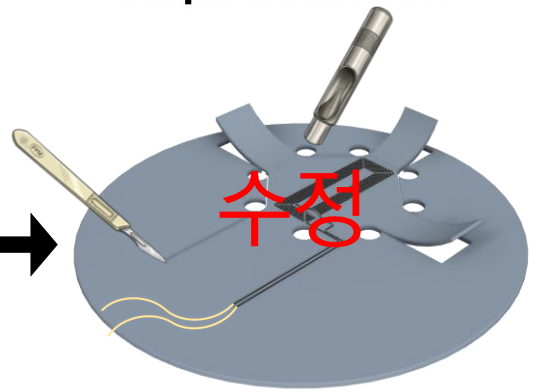
Step2 EGaln DIW



Step3 Bridging



Step4 Covering




Step5 Wire Inserting / Cutting

추가

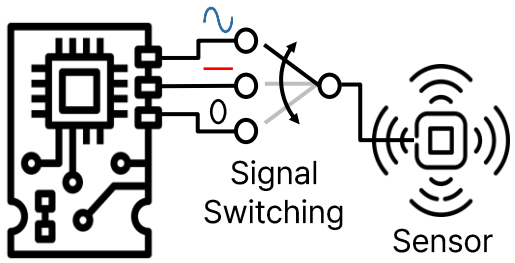
Step6 돌기

# III. Measurement System

# III. Measurement System

## 3.1 Time Division Measurement (TDM) Architecture

### Time Division Measurement (TDM)




**<Signal Acquisition Timeline>**



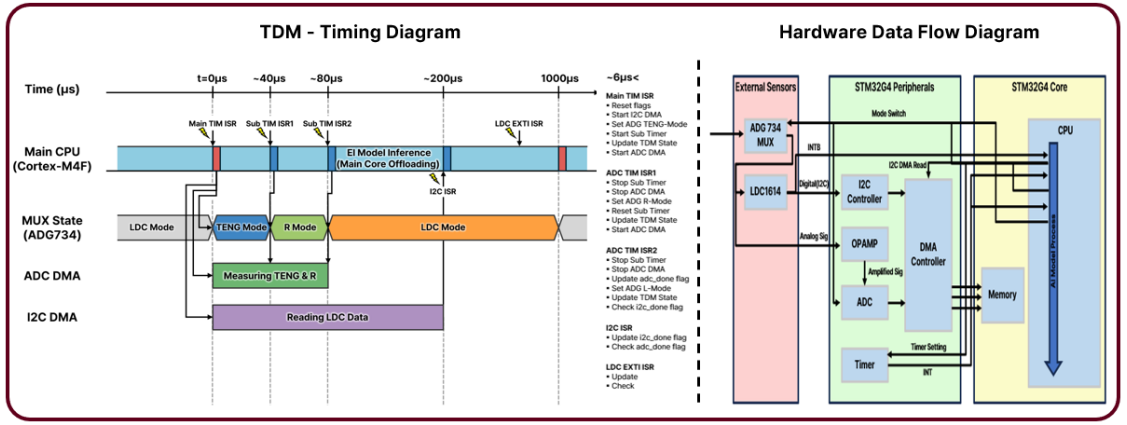

AC : $f = \frac{1}{2\pi\sqrt{LC}}$

DC : $R$

— : $V$

- 시분할 측정 방식을 통해 단일 센서에 순차적으로 AC excitation, DC excitation, No excitation을 반복하여 인가
- 각 상태에서 센서의 **공진주파수**, **저항**, **전압**(TENG)을 각각 독립적으로 측정

**Timer Based Parallel Non-Blocking DMA Processing for Embedded AI**



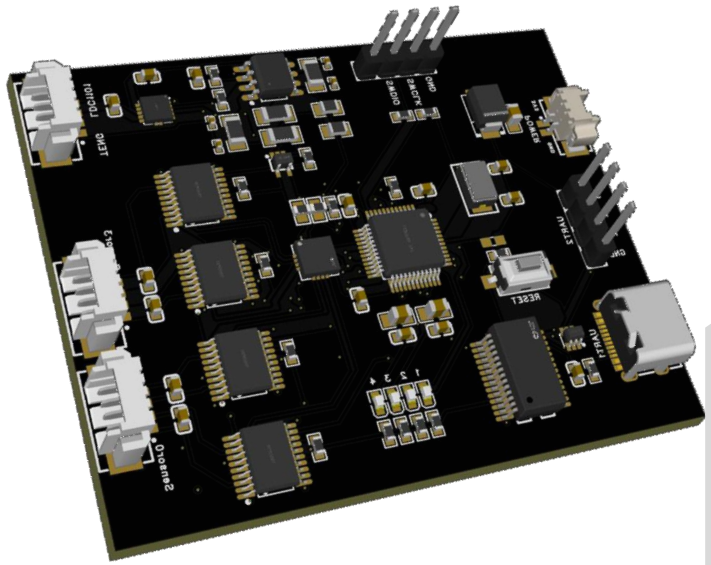

* TDM : Time Division Measurement

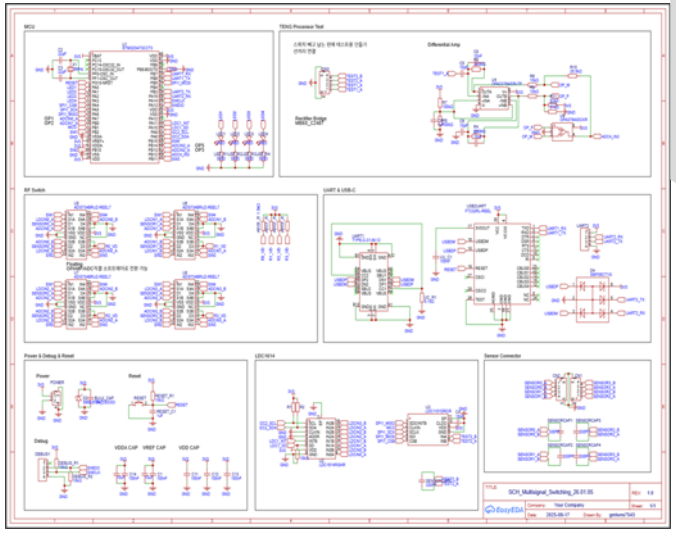
* Sensing Period : 1kHz
* CPU Idle Ratio : 93.908%

1. **Minimized overhead** : Implemented using an event-driven approach combining DMA, EXTI, and timer interrupts
2. **Fixed sampling rate** : Prevents jitter when interfacing with AI models

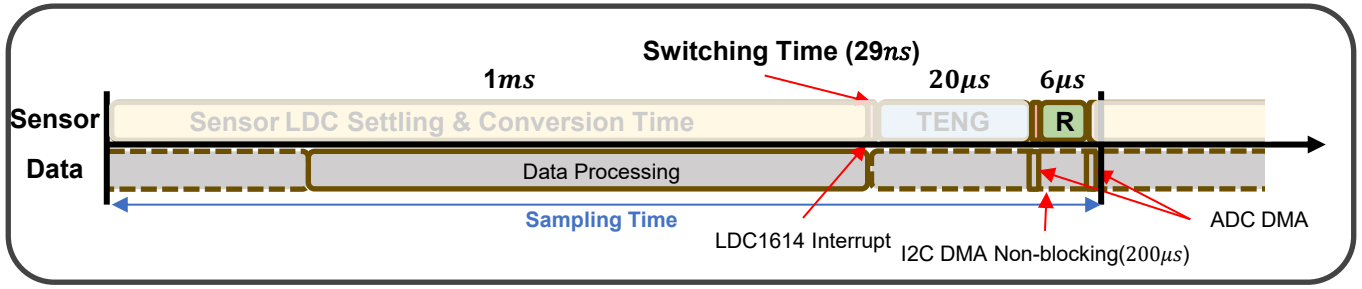
# III. Measurement System

## 3.2 Multi-Signal Switching PCB





## Circuit Working Mechanism




$L$

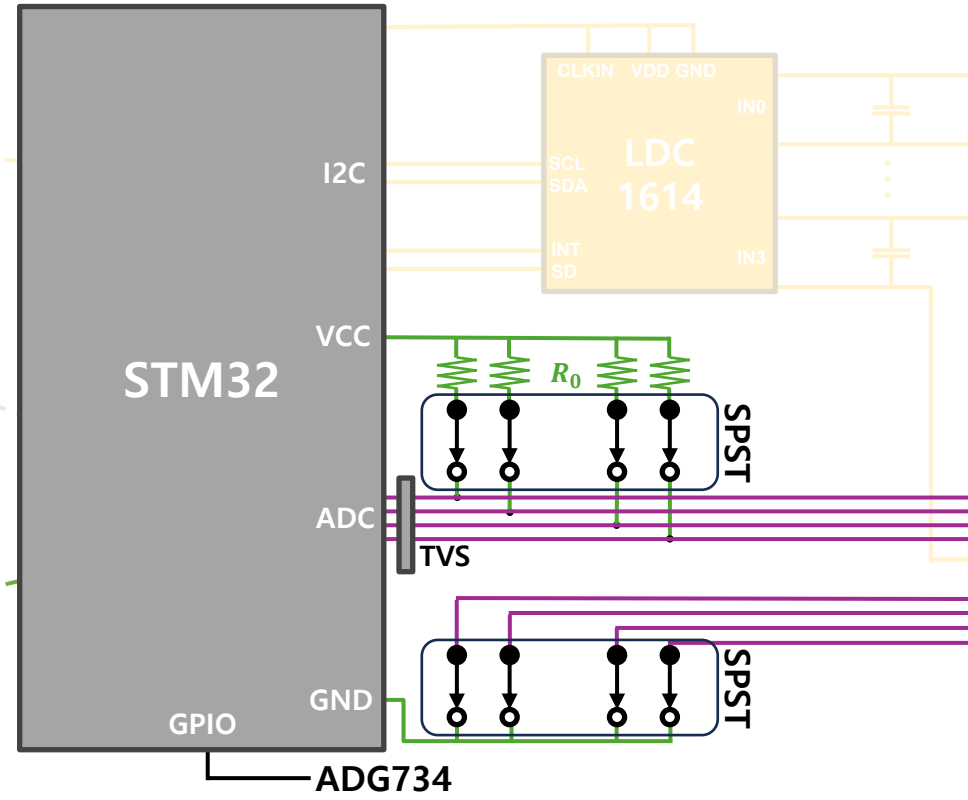
$$28bit = \frac{1}{2\pi\sqrt{LC}} / f_{ref}$$

$V_{TENG}$

$$\mathrm{ADC} = V \times Gain_{op}$$

$R_{DC}$

$$\mathrm{ADC} = \frac{R}{R_0 + R} \times Gain_{op}$$

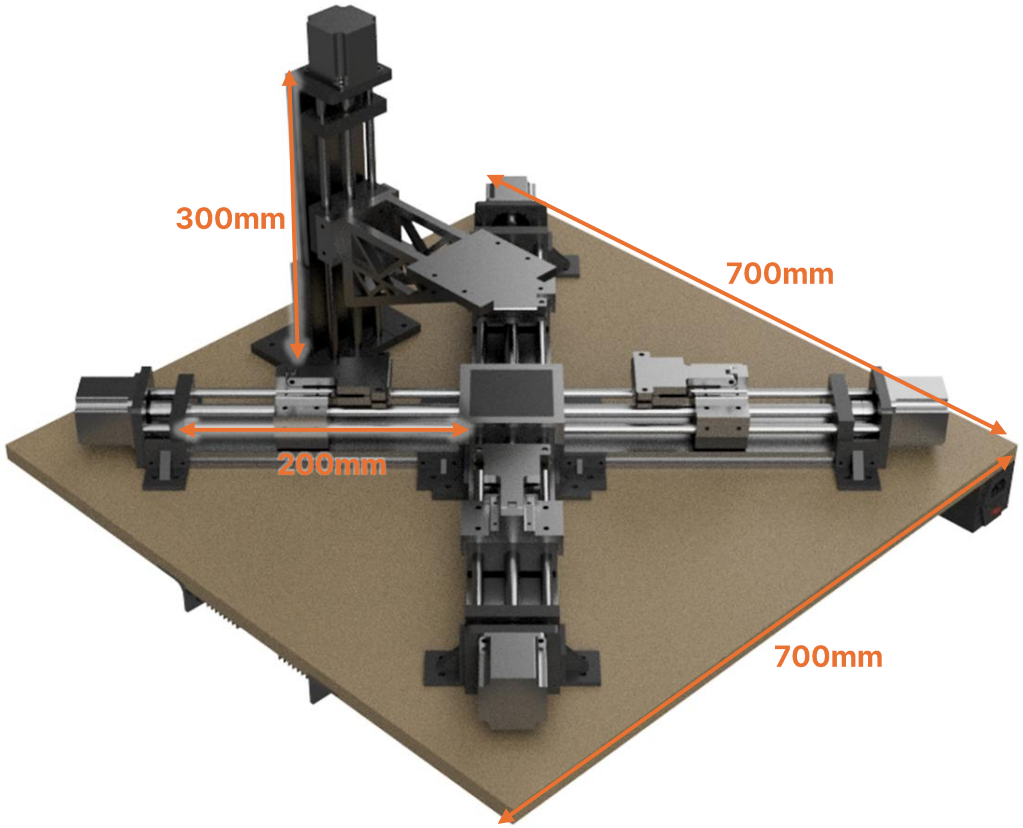


# IV. Signal Characterization

# IV. Signal Characterization

## 4.1 Experimental Setup

### Five-Axis Test Platform




Fabrication of a test platform for data acquisition

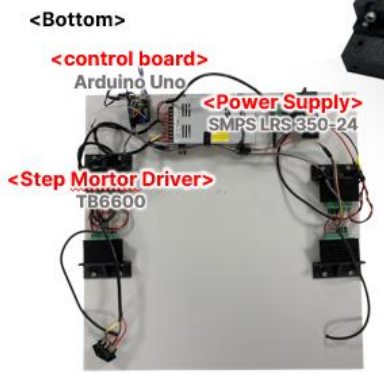








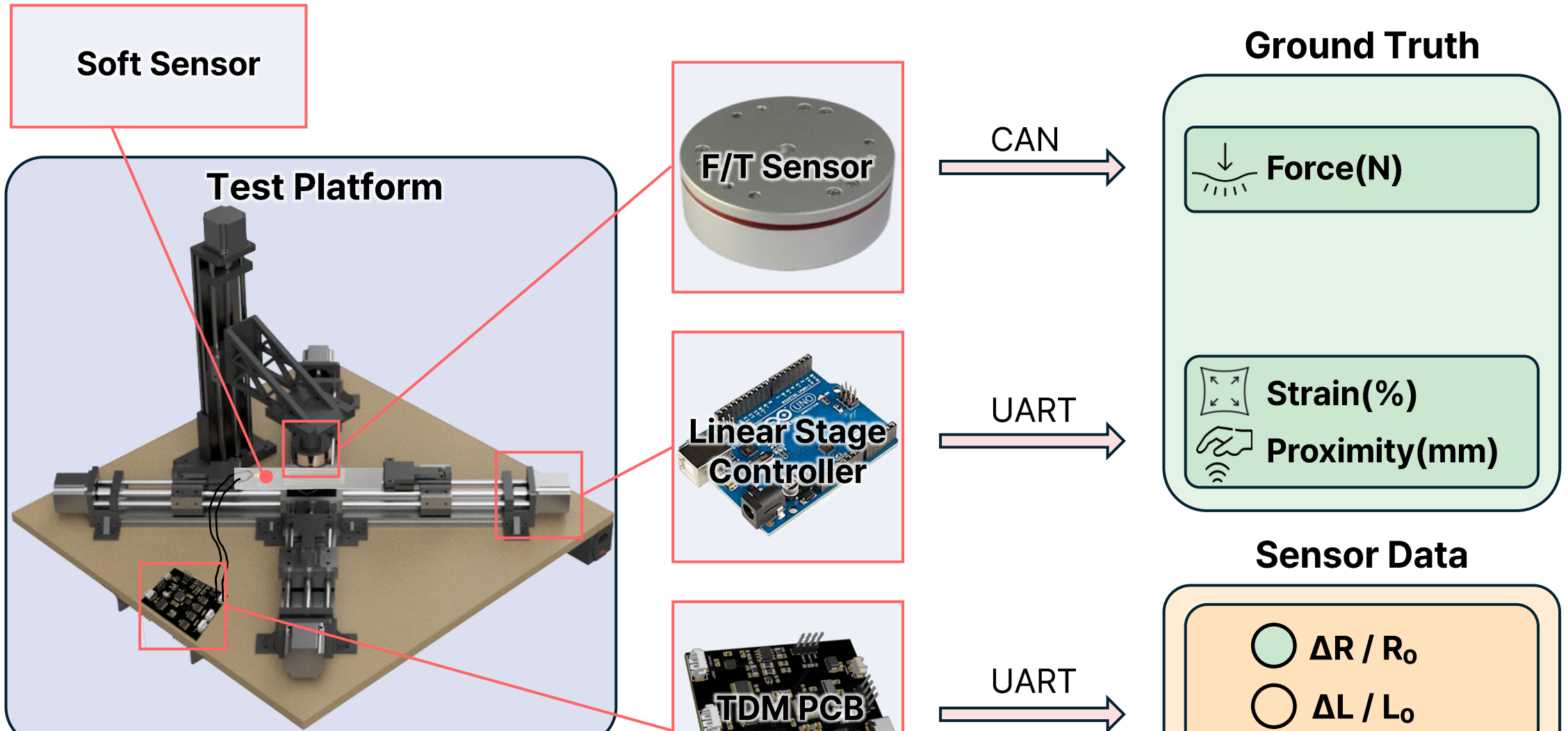


# IV. Signal Characterization

## 4.1 Experimental Setup

### Synchronized Acquisition

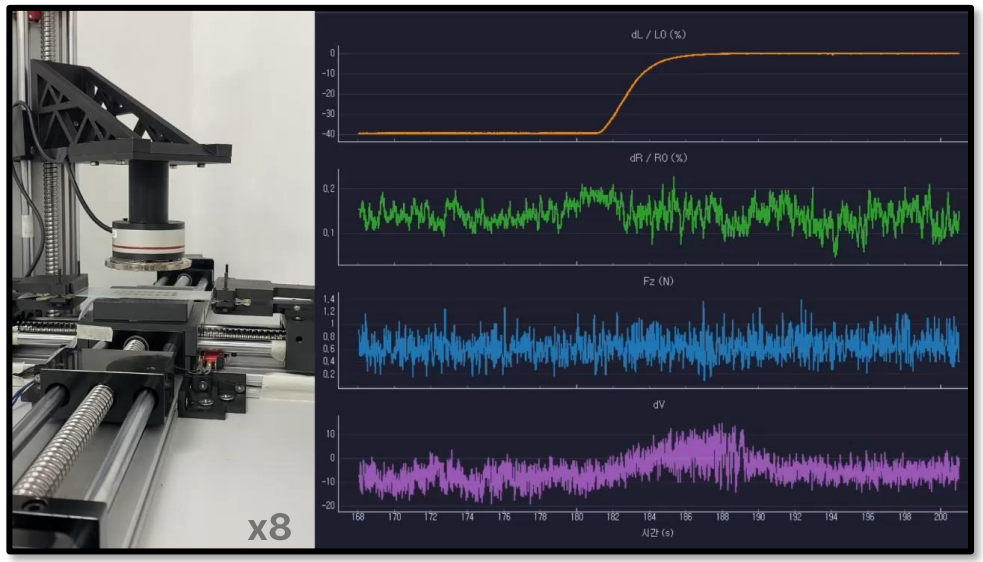


# IV. Signal Characterization

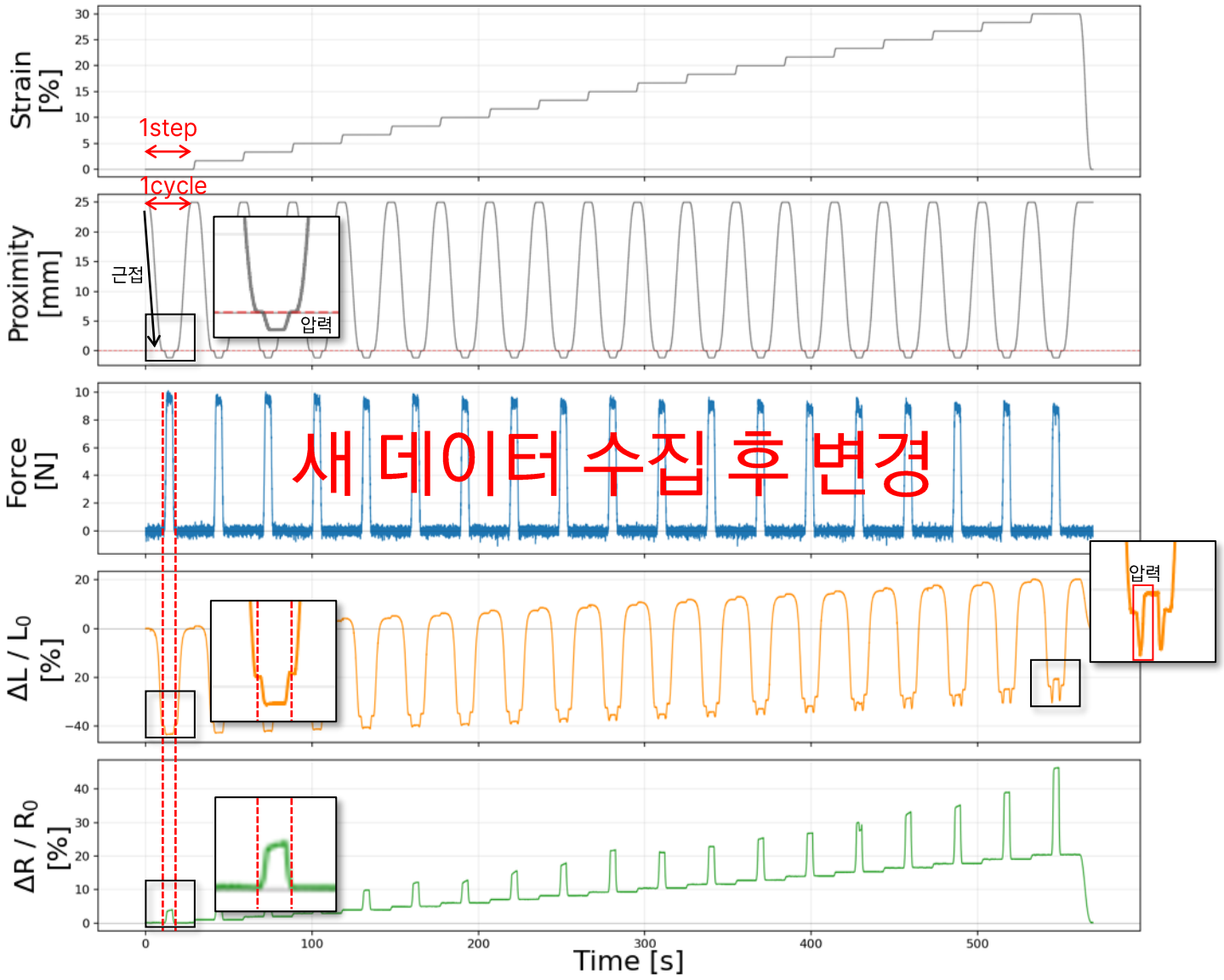
## 4.2 Data Acquisition

**1. Strain(discrete)**
- *0mm(0%) → 36mm(30%) :* 총 19 steps (*2mm* 간격)

**2. Position(continuous)**
- 근접 + 압력
- 근접: *25mm ~ 0mm*
- 압력: *0mm ~ -1.2mm (0N~9N)*




▲ 실측 동영상 (8 배속)

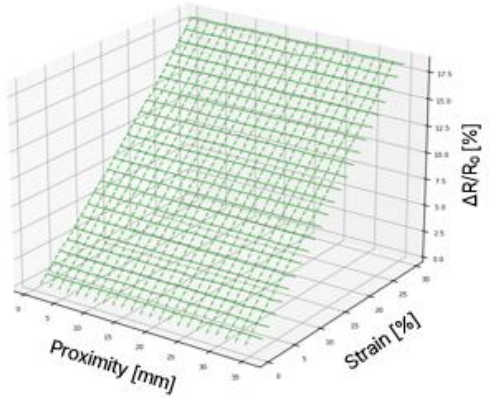


# IV. Signal Characterization

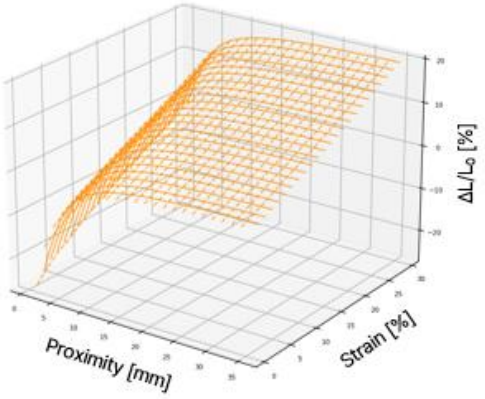
## 4.3 Non-Contact Mode: Stain-Proximity

- **3D Plot**

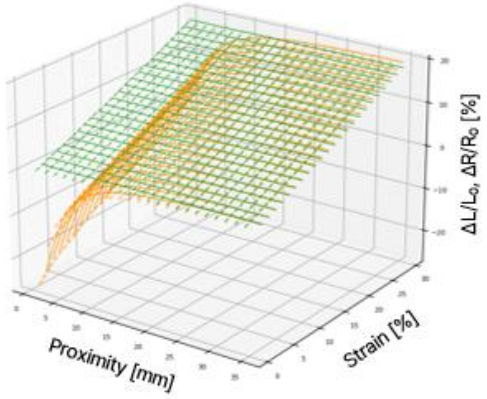
Resistance




Inductance

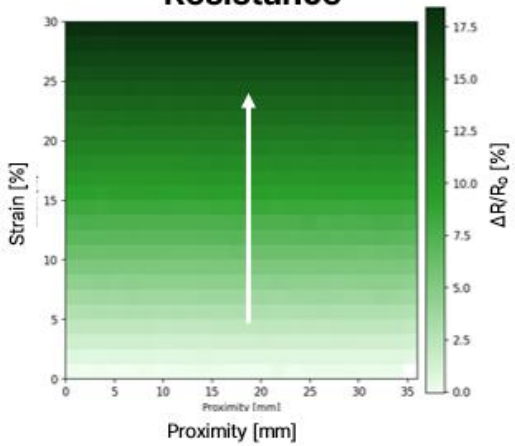



Inductance + Resistance

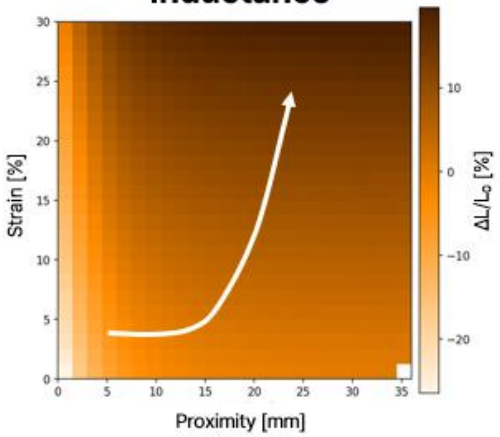



- **2D Heatmap & Contour**

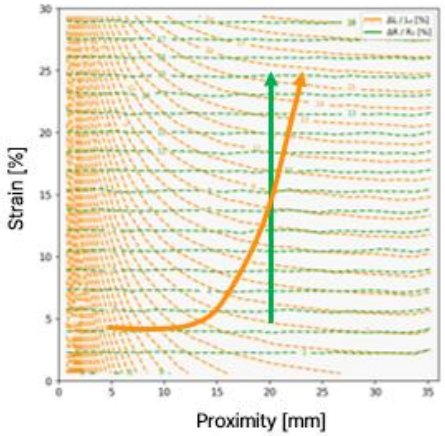
Resistance




Inductance




Contour

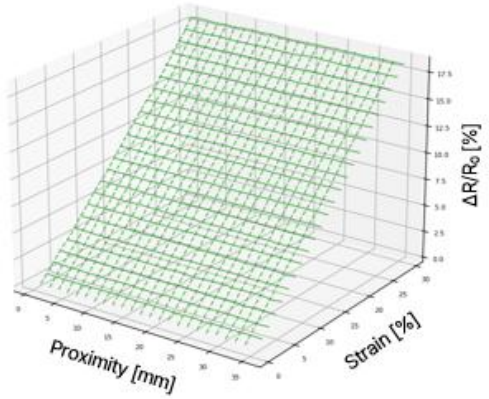


# IV. Signal Characterization

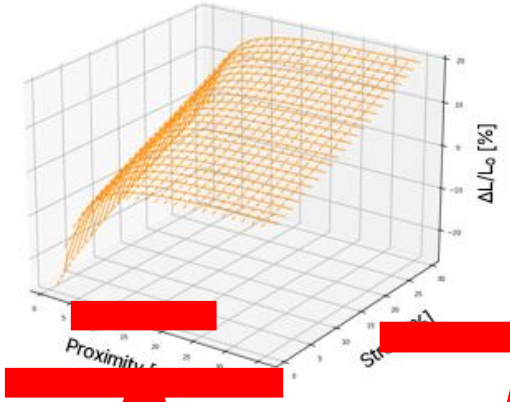
## 4.4 Contact Mode: Stain-Pressure

- **3D Plot**

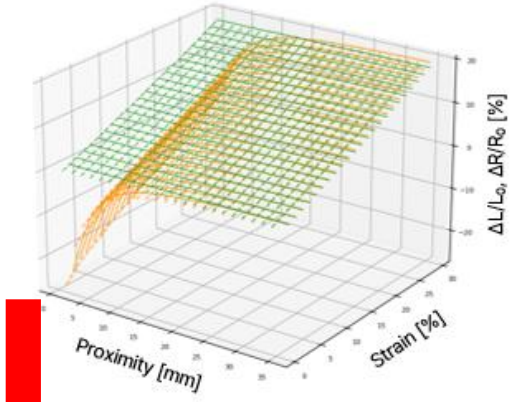
Resistance




Inductance

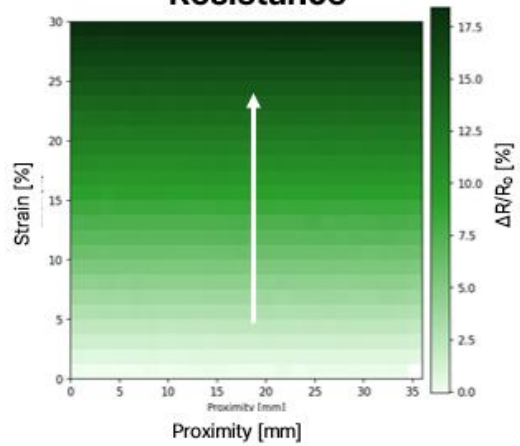



Inductance + Resistance

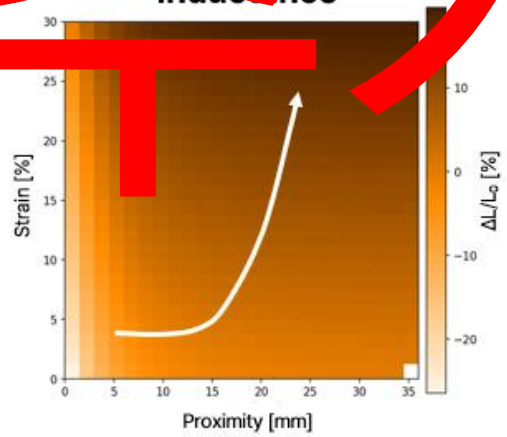



- **2D Heatmap & Contour**

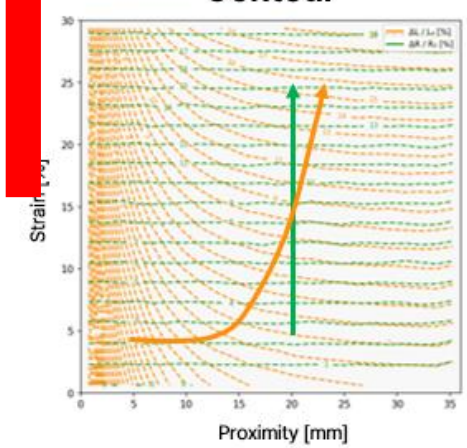
Resistance




Inductance



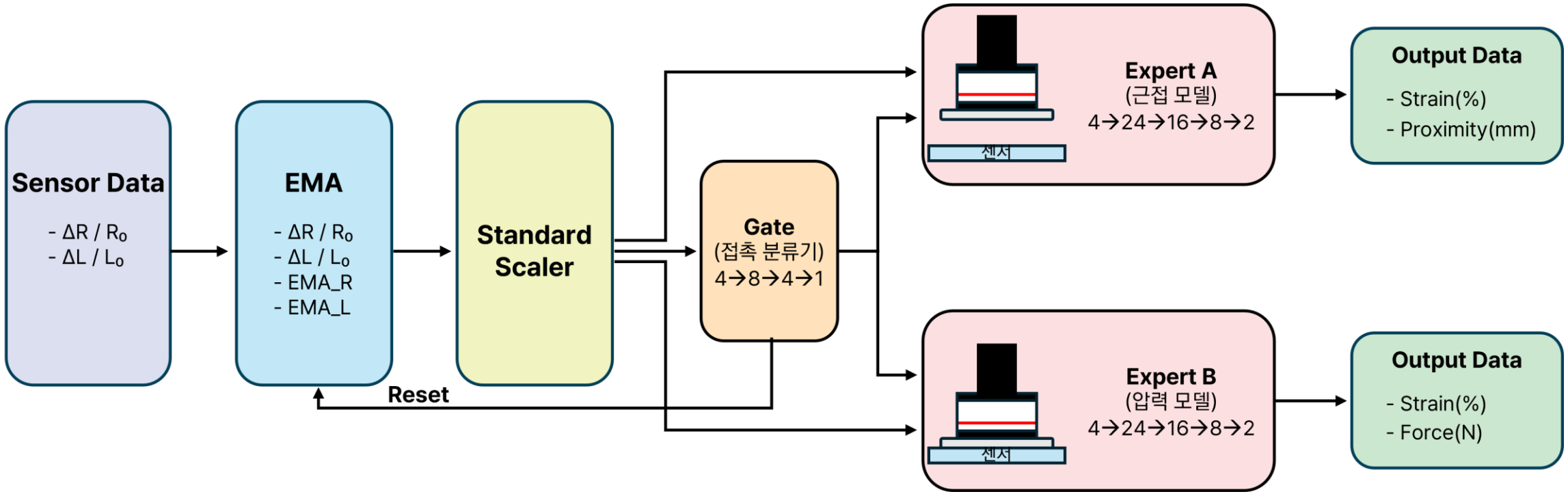

Contour



# V. Decoupling Model and  Embedded Implementation for Real-Time Inference

## 5.1 Decoupling Model

- SLS-EMA + Gate-MOE (Mixture of Experts) 모델

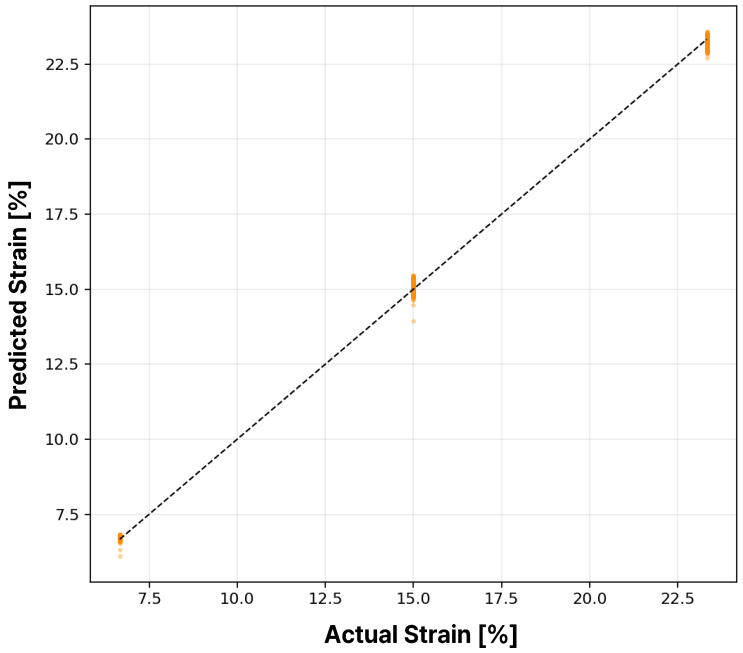


# V. Decoupling Model and Embedded Implementation for Real-Time Inference

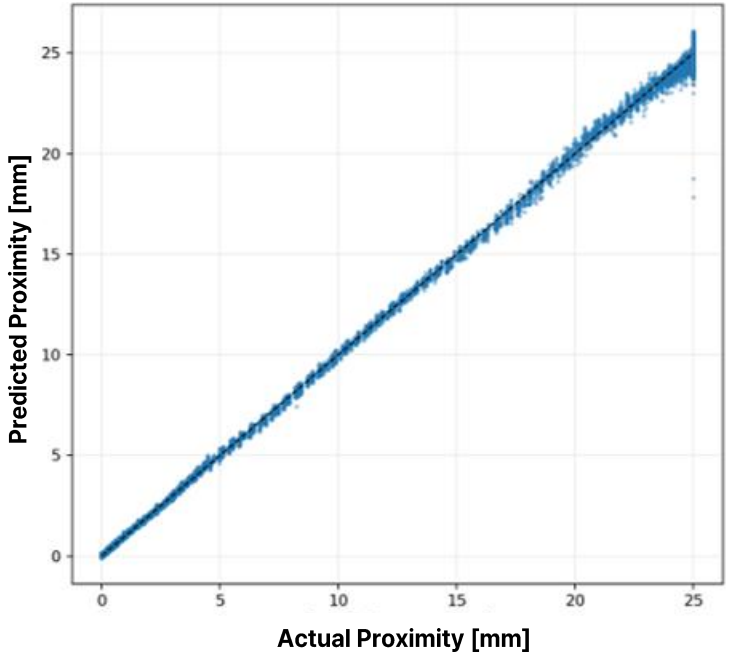
## 5.1 Decoupling Model

### - Result

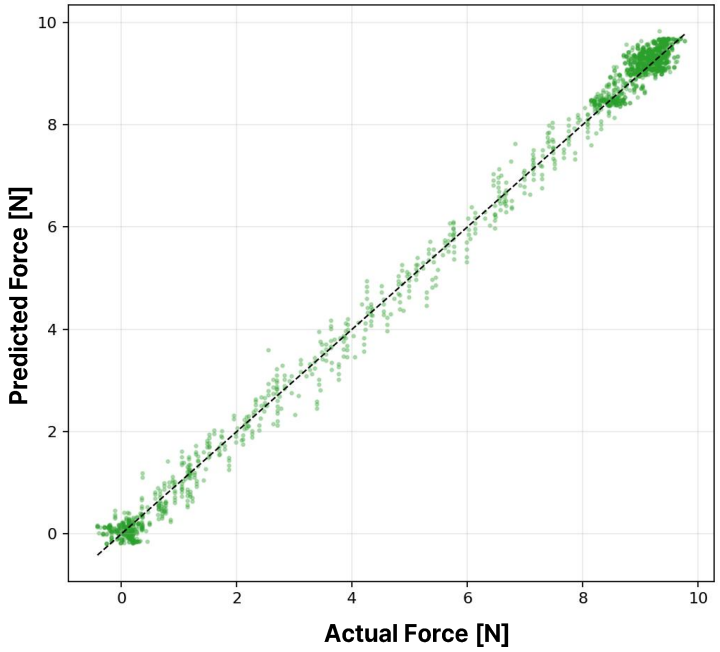
<Strain>




<Proximity>



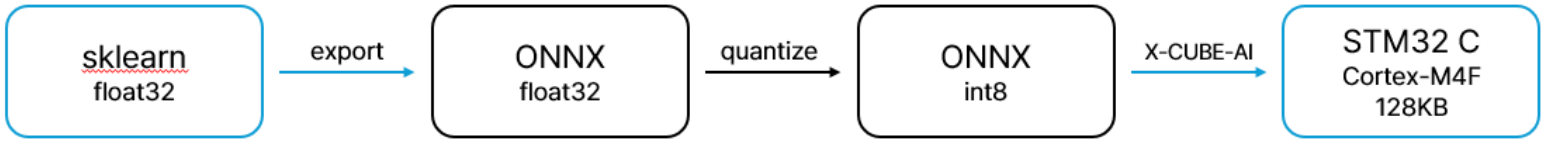

<Force>




| 모델 | 게이트 정확도 | Strain $R^2$ | Strain RMSE | Proximity $R^2$ | Proximity RMSE | Force $R^2$ | Force RMSE |
|---|---|---|---|---|---|---|---|
| Gate-MOE SLS-EMA 모델 | 1.000 | 0.999 | 0.108%p | 0.999 | 0.318mm | 0.995 | 0.253N |

## 5.2 Embedded Implementation

- 배포 파이프라인




- 모델 파라미터 최적화 → 메모리 사용량, 연산량, 추론속도 개선
  - Flash: 112KB → 978B
  - MACs: 50,306 → 892
  - 연산속도: 1068.78us → 141.01us

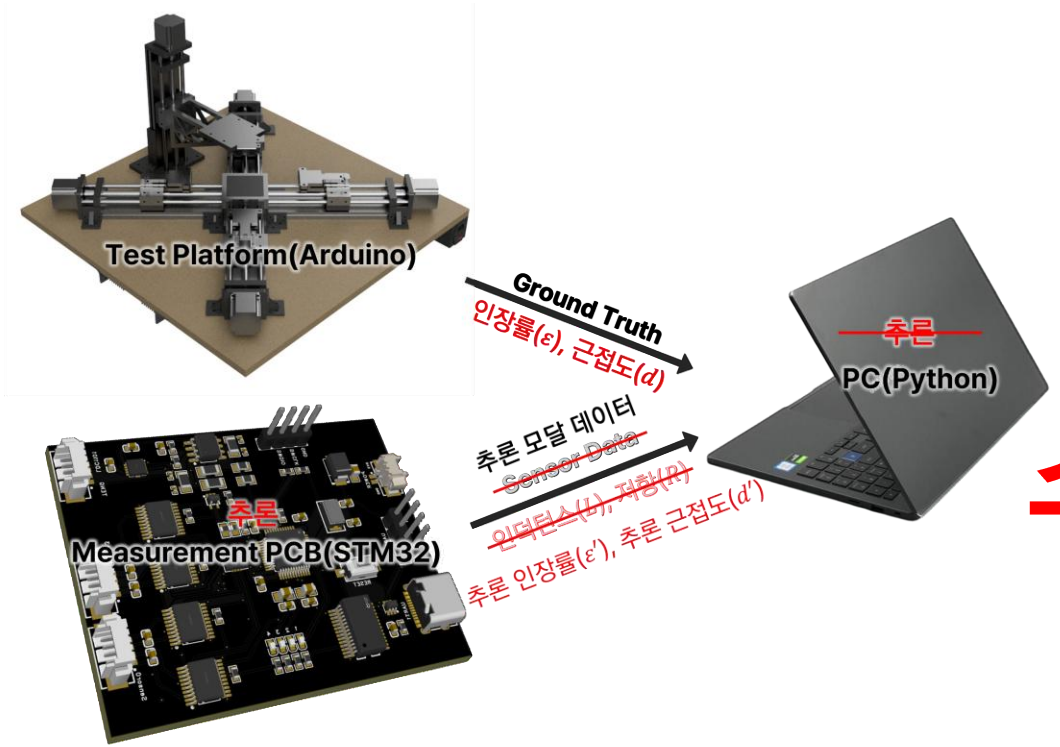
- 임베딩 최적화
  1. 컴파일러 최적화
     - Og → 57KB
  2. X-CUBE-AI 런타임
     - SMLAD DSP, Word Packing
  3. 모델 양자화
     - 50,306파라미터 Flash 196KB->52KB
     - 양자화 오차 → eps-MAE: 0.061%, d-MAE: 0.070mm

| S1 구조 | S2 구조 | 파라미터 | INT8 FLASH | MACS | MAE ε (%) | MAE D (mm) | MAE D≤15 (mm) |
|---|---|---|---|---|---|---|---|
| 1→4→1 | 2→8→1 | 46 | 46 B | 32 | 0.3810 | 2.1149 | 0.8058 |
| 1→8→1 | 2→16→1 | 90 | 90 B | 64 | 0.3630 | 2.1102 | 0.7810 |
| 1→8→1 | 2→32→1 | 154 | 154 B | 112 | 0.3680 | 1.9779 | 0.5610 |
| 1→4→4→1 | 2→8→8→1 | 138 | 138 B | 112 | 0.3518 | 1.8368 | 0.3943 |
| 1→16→1 | 2→16→8→1 | 242 | 242 B | 200 | 0.3514 | 1.8016 | 0.3476 |
| 1→16→8→1 | 2→16→8→1 | 370 | 370 B | 320 | 0.3493 | 1.7978 | 0.3463 |
| 1→16→8→1 | 2→32→16→1 | 818 | 818 B | 744 | 0.3491 | 1.7822 | 0.3217 |
| 1→8→1 | 2→32→16→1 | 666 | 666 B | 608 | 0.3496 | 1.7850 | 0.3258 |
| 1→16→8→1 | 2→64→32→1 | 2,482 | 2.4 KB | 2,360 | 0.3491 | 1.7813 | 0.3165 |
| 1→32→16→1 | 2→64→32→1 | 2,914 | 2.8 KB | 2,768 | 0.3519 | 1.7792 | 0.3118 |
| 1→16→8→4→1 | 2→32→16→8→1 | 978 | 978 B | 892 | 0.3515 | 1.7651 | 0.3222 |
| 1→16→8→1 | 2→128→64→1 | 8,882 | 8.7 KB | 8,664 | 0.3497 | 1.7735 | 0.3177 |
| 1→64→32→1 | 2→128→64→1 | 10,946 | 10.7 KB | 10,656 | 0.3490 | 1.7730 | 0.3133 |
| 1→128→128→64→1 | 2→128→128→64→1 | 50,306 | 52KB | 49,664 | 0.3435 | 1.7830 | 0.3009 |

## 5.2 Embedded Implementation

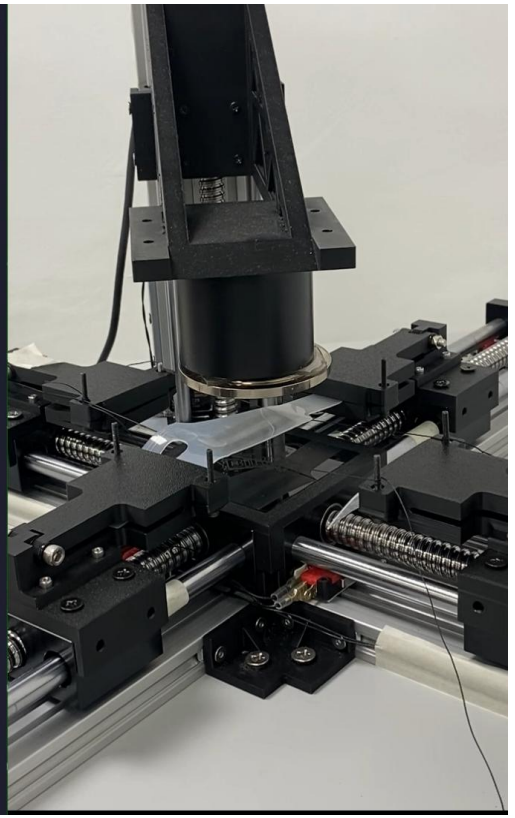
## 5.3 Real-Time Validation

### EI-based Realtime Decoupling




수정

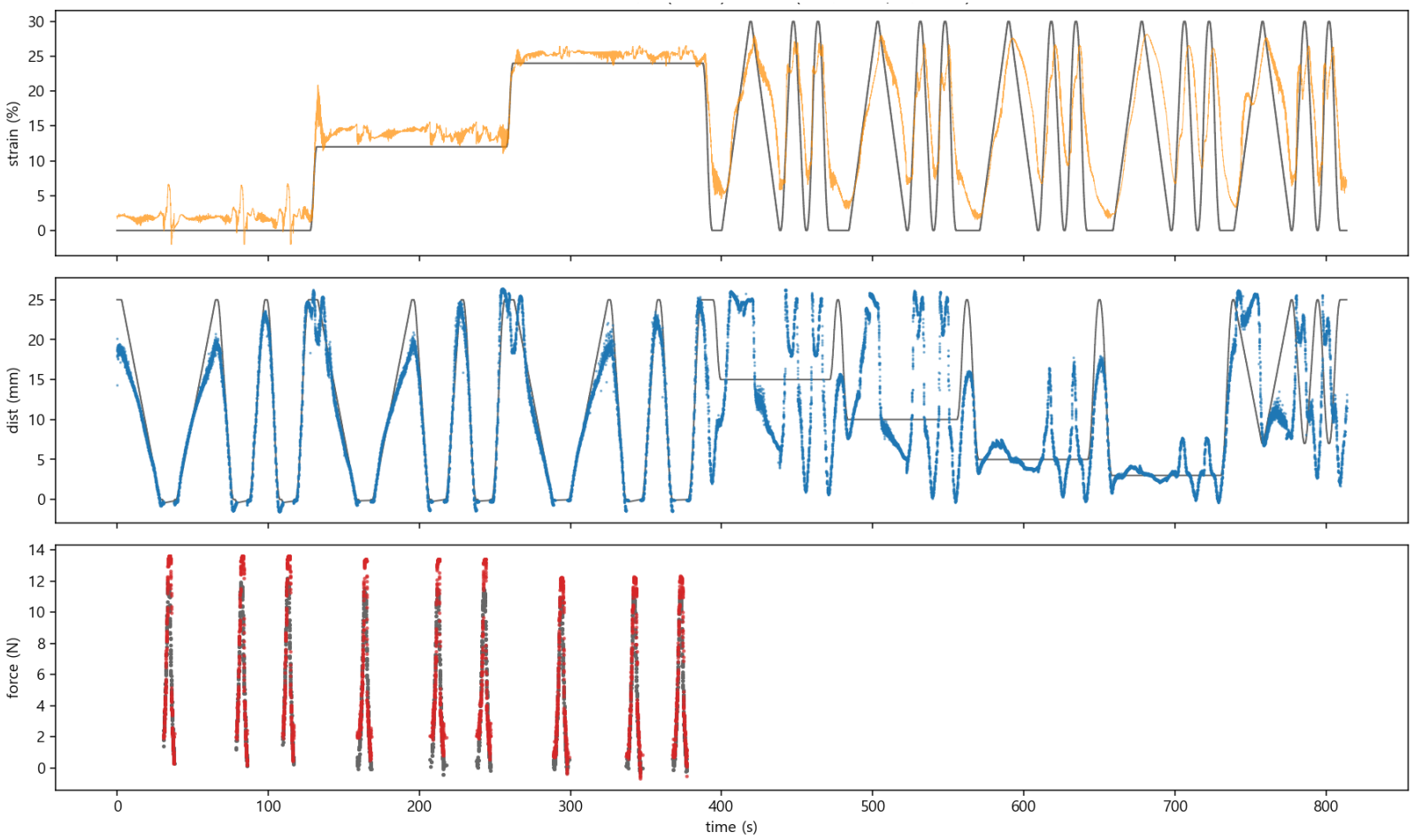


# V. Decoupling Model and Embedded Implementation for Real-Time Inference

## 5.3 Real-Time Validation

### EI-based Realtime Decoupling



# VI. Conclusion

# VI. Conclusion

## 6.1 Summary

# VI. Conclusion

## 6.2 Future Work

# Thank you

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